

Project Idea: SaniShield AI

Climate-Resilient Smart Sanitation Network for Northern Ghana

Core Idea

SaniShield AI is a low-cost, AI-powered sanitation monitoring and response system designed for flood- and drought-prone communities in Northern Ghana.

The system combines:

- Smart IoT sanitation sensors
- Offline-first mobile technology
- AI flood-risk prediction
- SMS emergency alerts
- GIS sanitation mapping
- Solar-powered infrastructure

to prevent toilet overflow, water contamination, and sanitation system collapse during climate disasters.

The BIG Problem You're Solving

During floods:

- Toilets overflow
- Waste contaminates water
- Schools shut down
- Disease spreads rapidly

During drought:

- Water-based toilets fail
- Hygiene becomes difficult
- Waste treatment systems break down

Most communities only react AFTER disaster happens.

Your innovation:

SaniShield AI predicts sanitation failures BEFORE they become disasters.

The Winning Angle

Most teams will likely build:

- simple reporting apps
- dashboards
- awareness platforms

Your project becomes stronger because it combines:

- ✓ AI
- ✓ Hardware
- ✓ Climate resilience
- ✓ Offline support
- ✓ Real-world deployment
- ✓ Child/school safety
- ✓ Data intelligence
- ✓ Emergency response
- ✓ Sustainability
- ✓ Low-cost deployment

This makes it look like a real UNICEF pilot-ready system.

How The System Works

1. Smart Toilet Monitoring Unit

Each toilet/sanitation point gets a low-cost monitoring device.

Hardware:

- Old Android phones OR ESP32
- Ultrasonic waste-level sensor

- Flood/water-level sensor
- Temperature/humidity sensors
- Solar charging module

The device monitors:

- Overflow risk
 - Floodwater intrusion
 - Unsafe gas buildup
 - Structural instability
 - Usage trends
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2. AI Prediction Engine

The AI analyzes:

- Rainfall patterns
- Historical flooding data
- Toilet fill levels
- Community sanitation activity

Then predicts:

- Which toilets may overflow
 - Which schools are at risk
 - Which areas may face contamination
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3. Offline-First Mobile App

A Flutter mobile app for:

- sanitation workers
- schools
- local assemblies
- NGOs

Features:

- Offline reporting

- Emergency alerts
- Maintenance requests
- GPS mapping
- Photo uploads
- Community incident reports

Works with:

- Low internet
- Weak signal
- Rural conditions

Syncs automatically when internet returns.

4. SMS Alert System

Critical because many rural users don't have smartphones.

When danger is detected:

- SMS alerts sent to sanitation officers
- Schools receive warnings
- Local maintenance teams notified

Example:

"Flood risk detected at Sakasaka Primary School toilet block. Overflow risk in 6 hours."

5. Smart GIS Risk Dashboard

A web dashboard for UNICEF/local government.

Shows:

- Live sanitation map
- High-risk zones
- Flood-prone toilets
- Maintenance status

- Disease-risk indicators
- Community reports

Built with:

- Next.js
 - Mapbox/Leaflet
 - Real-time analytics
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6. Climate-Resilient Infrastructure Layer

This is VERY important for judging.

You're not just building software.

You're proposing:

- Elevated toilet foundations
 - Flood-resistant biodigester systems
 - Solar-powered monitoring
 - Water-efficient sanitation
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Why This Could Win

1. It Solves The FULL Sanitation Chain

Most ideas solve one part.

Yours addresses:

- Monitoring
 - Prevention
 - Maintenance
 - Response
 - Data collection
 - Climate resilience
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2. UNICEF LOVES Preventive Systems

Instead of:

“Let’s clean after flooding”

You say:

“Let’s predict and prevent sanitation disasters before children are affected.”

That is a BIG difference.

3. Strong Real-World Feasibility

You can prototype this cheaply.

Prototype stack:

- ESP32
- Cheap sensors
- Old Android phones
- Flutter app
- Firebase/Supabase
- Next.js dashboard

This makes judges believe it can actually deploy.

Tech Stack

Mobile App

- Flutter

Backend

- Node.js / Express OR NestJS

AI/Prediction

- Python
- Scikit-learn
- TensorFlow Lite

Dashboard

- Next.js

Database

- PostgreSQL or Firebase

Hardware

- ESP32
 - Ultrasonic sensor
 - Water level sensor
 - Solar power
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AI Features (Important for Winning)

Predictive Overflow Detection

Uses historical fill-rate patterns.

Flood Risk Intelligence

Combines:

- weather forecasts
- rainfall data
- local flood history

Maintenance Optimization

AI suggests:

- best waste collection routes
 - priority repair zones
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UNICEF Impact Alignment

Your project directly supports:

- Child health
 - School sanitation
 - Climate adaptation
 - Disease prevention
 - Rural resilience
 - SDGs
 - Smart communities
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Business/Sustainability Model

This matters A LOT.

Revenue/Long-Term Sustainability

Offer system as:

- Government sanitation infrastructure service
- NGO deployment package
- District assembly monitoring platform

Low-cost subscription for monitoring.

Demo Prototype For Hackathon

You do NOT need full deployment.

Build:

- ✓ One sensor prototype
- ✓ Flutter app mockup
- ✓ Live dashboard
- ✓ Simulated flood alert
- ✓ SMS notification demo
- ✓ AI prediction visualization

That alone would look VERY strong.

Killer Pitch Line

“SaniShield AI helps Northern Ghana move from reacting to sanitation disasters to predicting and preventing them before children are put at risk.”

Bonus Innovation (VERY Strong)

Add:

Community Voice Reporting

People can report sanitation issues via:

- local language voice notes
- SMS
- missed calls

This helps:

- low literacy communities
 - elderly users
 - remote villages
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Suggested Team Roles

- AI/ML Engineer
- Frontend Developer
- Backend Developer
- Hardware/IoT Engineer
- UI/UX Designer
- Pitch/Research Lead